

Movie Rental Data Warehouse Design
From OLTP Schema to Dimensional Model and ETL Design
Data Warehousing / Data Architecture

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1. Introduction

The movie rental OLTP database is designed for daily operational transactions such as customer registration, film rentals, payments, inventory management, and staff activities. Although the OLTP system is efficient for transaction processing, it is not optimized for analytical reporting and business intelligence analysis.

Business managers need a centralized Data Warehouse that can support analytical queries, reporting, trend analysis, and decision-making. A dimensional data warehouse improves query performance and simplifies complex operational data into an analytical structure.

This project proposes a Movie Rental Data Warehouse based on dimensional modeling principles using fact tables, dimension tables, and ETL processes. The warehouse supports business analysis related to rental activity, customer behavior, film popularity, revenue generation, store performance, and time-based trends.

The project also includes an ETL pipeline implemented using Python, Pandas, and MySQL. The ETL process extracts data from the OLTP Sakila database, transforms and cleans the data, and loads it into a dimensional warehouse schema designed for analytics and reporting.

Additionally, analytical charts and dimensional model diagrams were created to support business intelligence visualization and reporting.

2. Business Questions

The warehouse is designed to answer the following business questions:

Which films are rented most frequently?

Which stores generate the highest revenue?

How does revenue change by month, quarter, or year?

These business questions help management analyze film popularity, store financial performance, and revenue trends over time in order to support business intelligence and decision-making.

3. Dimensional Model Design

3.1 Selected Business Processes

Business Process	Why Selected	Business Process
Rental Transactions	Core business activity of the system.	Rental analysis and late return tracking.
Payment Transactions	Main source for revenue analysis.	Revenue and payment analytics.
Inventory Availability	Supports inventory monitoring.	Inventory availability statistics.

3.2 Fact Tables

Fact Table	Grain	Measures	Related Dimensions
Fact_Rental	One row per rental transaction	rental_count, rental_duration_days, late_return_flag	Dim_Date, Dim_Customer, Dim_Film, Dim_Store, Dim_Staff
Fact_Payment	One row per payment transaction	payment_amount, payment_count	Dim_Date, Dim_Customer, Dim_Film, Dim_Store, Dim_Staff
Fact_Inventory	One row per inventory item per store	inventory_count, available_flag	Dim_Film, Dim_Store

3.3 Dimension Tables

Dimension Table	Main Attributes	Purpose
Dim_Date	date_key, full_date, month, year	Time analysis
Dim_Customer	customer_key, full_name, city, country	Customer analysis
Dim_Film	film_key, film_title, rental_rate	Film analysis
Dim_Store	store_key, store_id, city	Store analysis
Dim_Staff	staff_key, full_name	Staff analysis
Dim_Category	category_key, category_name	Category analysis
Dim_Actor	actor_key, full_name	Actor analysis

3.4 Bridge Tables

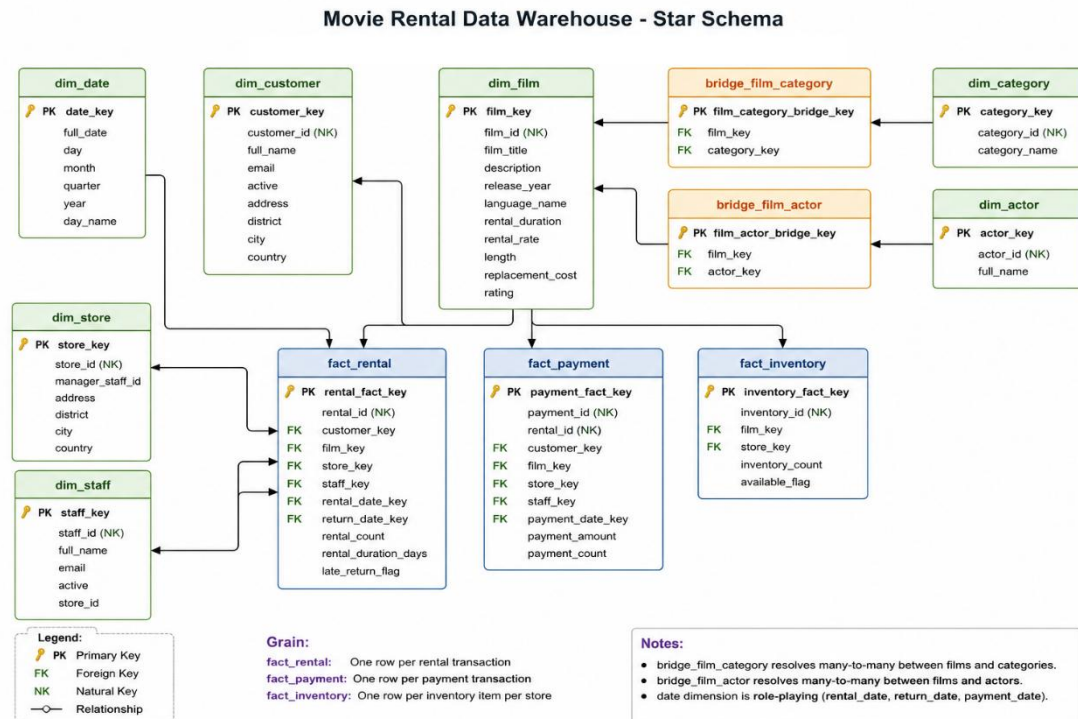
Bridge Table	Keys	Purpose
Bridge_Film_Category	film_key, category_key	Handles film-category relationships
Bridge_Film_Actor	film_key, actor_key	Handles film-actor relationships

3.5 Schema Type

The proposed warehouse follows a hybrid star schema design that combines traditional star-schema principles with bridge tables to handle many-to-many relationships between films, categories, and actors.

4. Dimensional Model Diagram

The following diagram illustrates the final dimensional model of the Movie Rental Data Warehouse. The model includes fact tables, dimension tables, and bridge tables used to support analytical reporting and business intelligence analysis.



5. ETL Design

The ETL pipeline was implemented using Python, Pandas, SQLAlchemy, and MySQL. The purpose of the ETL process is to extract operational data from the Sakila OLTP database, transform it into a dimensional model, and load it into the Movie Rental Data Warehouse.

5.1 Extract

Source OLTP Table	Used For
customer	Building Dim_Customer
address	Adding customer and store address details
city	Adding city information
country	Adding country information
film	Building Dim_Film and film-related facts
language	Adding film language information
category	Building Dim_Category
film_category	Building Bridge_Film_Category
actor	Building Dim_Actor
film_actor	Building Bridge_Film_Actor
inventory	Building Fact_Inventory and connecting films to stores
rental	Building Fact_Rental
payment	Building Fact_Payment
staff	Building Dim_Staff
store	Building Dim_Store

5.2 Transform

Transformation	Description
Drop unnecessary columns	Removed operational columns such as <code>last_update</code> where not needed.
Create full names	Combined <code>first_name</code> and <code>last_name</code> for customers, staff, and actors.
Generate surrogate keys	Created warehouse keys such as <code>customer_key</code> , <code>film_key</code> , <code>staff_key</code> , <code>category_key</code> , and <code>actor_key</code> .
Create date dimension	Built <code>dim_date</code> from rental, return, and payment dates.
Build dimensions	Created customer, film, store, staff, category, actor, and date dimensions.
Build bridge tables	Created bridge tables for film-category and film-actor relationships.
Calculate rental duration	Calculated <code>rental_duration_days</code> from rental and return dates.
Detect late returns	Created <code>late_return_flag</code> by comparing actual rental duration with expected rental duration.
Create measures	Created measures such as <code>rental_count</code> , <code>payment_count</code> , <code>payment_amount</code> , <code>inventory_count</code> , and <code>available_flag</code> .
Save CSV outputs	Exported transformed warehouse tables as CSV files inside <code>transformed_data</code> .

5.3 Load

The loading stage loads the transformed tables into the MySQL data warehouse database named `movie_rental_warehouse`.

1	<code>dim_date</code>
2	<code>dim_customer</code>
3	<code>dim_film</code>
4	<code>dim_category</code>
5	<code>dim_actor</code>
6	<code>dim_store</code>
7	<code>dim_staff</code>
8	<code>bridge_film_category</code>
9	<code>bridge_film_actor</code>
10	<code>fact_rental</code>
11	<code>fact_payment</code>
12	<code>fact_inventory</code>

The dimension tables are loaded before the fact tables so that fact tables can reference existing surrogate keys. Bridge tables are also loaded before facts to support many-to-many relationships between films, categories, and actors.

6. Data Quality Considerations

Several data quality considerations were addressed during the ETL process to improve analytical accuracy and consistency within the Movie Rental Data Warehouse.

Data Quality Issue	Solution Applied
Duplicate operational columns	Removed unnecessary columns such as last_update.
Missing full names	Combined first and last names into unified full-name attributes.
Inconsistent date formats	Converted dates into standardized datetime formats.
Missing analytical structure	Created surrogate keys for all dimensions and fact tables.
Many-to-many relationships	Implemented bridge tables for film-category and film-actor relationships.
Rental duration inconsistency	Calculated rental duration using rental and return dates.
Late rental detection	Created late_return_flag to identify delayed returns.
Data integration complexity	Combined multiple OLTP tables into unified analytical dimensions.

Data Quality Summary

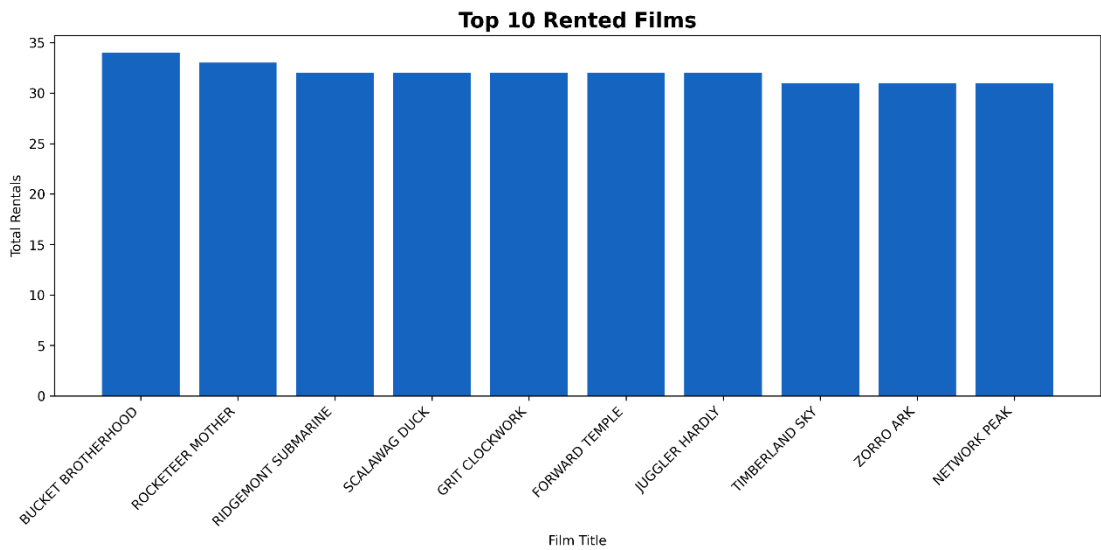
The ETL pipeline improves data consistency, dimensional integration, analytical usability, and reporting quality. The final warehouse structure supports reliable business intelligence analysis and simplifies complex operational relationships.

7. Charts and Analytics

The following charts were generated from the Movie Rental Data Warehouse to support analytical reporting and business intelligence analysis

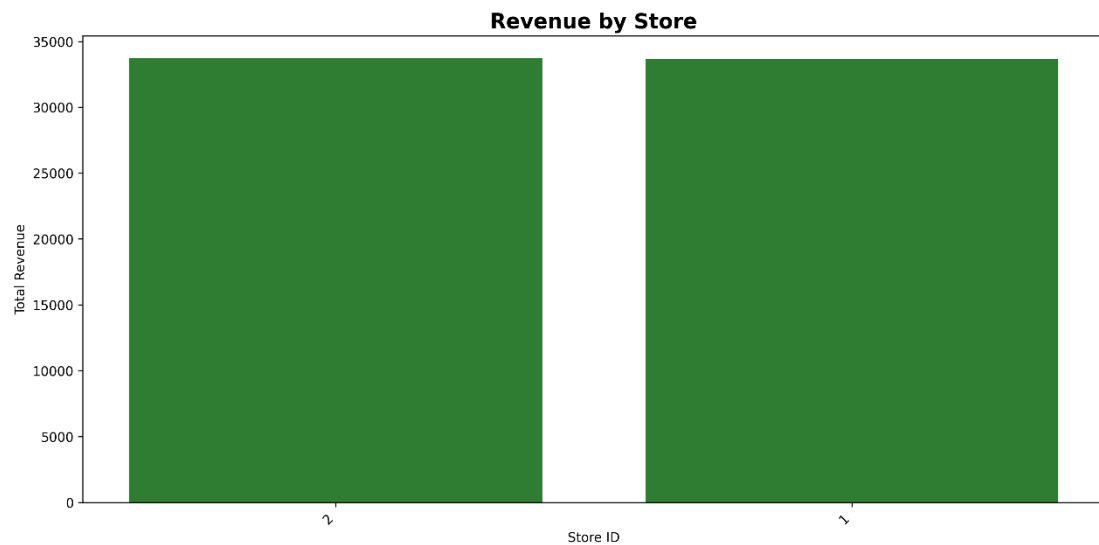
7.1 Top Rented Films

This chart shows the films with the highest rental activity. It helps identify the most popular films and customer preferences.



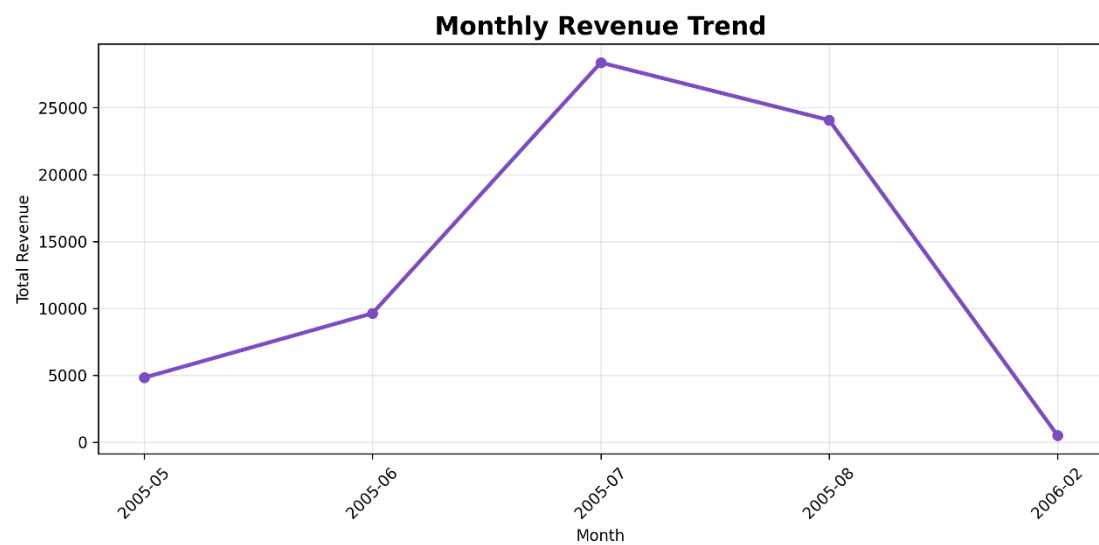
7.2 Revenue by Store

This chart compares total revenue generated by each store. It supports store performance and financial analysis.



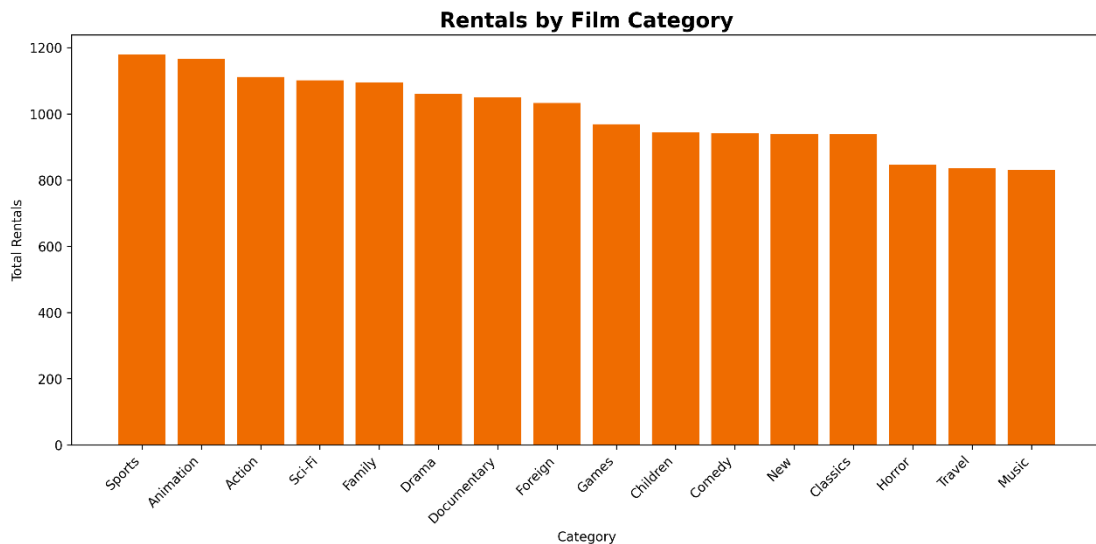
7.3 Monthly Revenue Trend

This chart illustrates how revenue changes over time by month. It helps analyze business growth and seasonal trends.



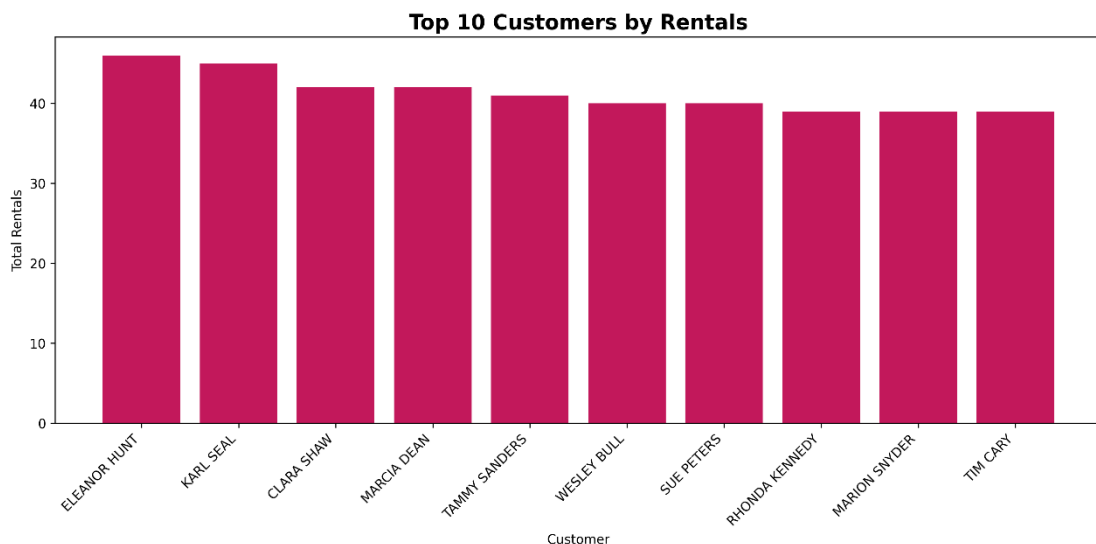
7.4 Rentals by Film Category

This chart displays rental activity by film category. It helps identify the most popular categories among customers.



7.5 Top Customers by Rentals

This chart identifies the customers with the highest rental activity. It supports customer behavior and loyalty analysis.



8. Conclusion

The proposed Movie Rental Data Warehouse successfully transforms the Sakila OLTP database into a dimensional analytical structure suitable for reporting and business intelligence analysis.

The warehouse design includes fact tables, dimension tables, and bridge tables to support rental analysis, revenue analysis, inventory tracking, customer behavior analysis, and time-based reporting.

The ETL pipeline extracts operational data, transforms it into analysis-ready structures, generates surrogate keys and analytical measures, and loads the final data into the warehouse. In addition, charts and dimensional diagrams were created to support visualization and decision-making.

The final dimensional model simplifies complex operational relationships and improves analytical performance, reporting capabilities, and business intelligence support.